How-To Library

Informatica

Deploy Data Quality on the AWS U.S. Intelligence Community Marketplace (10.5.1.1)

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Abstract

You can deploy Data Quality on the Amazon Web Services (AWS) U.S. Intelligence Community Marketplace. This deployment reference includes step-by-step instructions for deploying Data Quality on the AWS U.S. Intelligence Community Marketplace. It also includes information on prerequisites and troubleshooting.

Supported Versions

- Data Quality 10.5.1.1

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Overview

Customers of Amazon Web Services and Informatica can launch a Data Quality deployment from the AWS U.S. Intelligence Community Marketplace to create an Informatica domain in the AWS cloud and use Data Quality functionality.

Automated reference deployments use AWS CloudFormation templates to launch, configure, and run the AWS compute, network, storage, and other services required to deploy a specific workload on AWS.

Intended audience

This document is for users who are responsible for deploying Data Quality on the AWS U.S. Intelligence Community Marketplace.

As a user with administrator privileges to deploy applications on AWS, we assume that you are familiar with AWS resources such as Cloud formation, VPC, EC2, S3, RDS, Internet gateway, NAT Gateway, Route table, Security group, and Elastic IP.

You must be familiar with Data Quality, and we recommend that you are familiar with the concepts of IP CIDR and public and private IP addresses.

For Data Quality documentation, see the Informatica Documentation Portal.

Specialized knowledge

Before you deploy Data Quality, we recommend that you become familiar with the required AWS service. See "Additional Resources" on page 15 for more information.

Costs and licenses

You are responsible for the cost of the AWS services used while running this deployment. There is no additional cost for using this deployment.

The AWS CloudFormation template for this deployment includes configuration parameters that you can customize. Some of these settings, such as instance type, will affect the cost of deployment. See the pricing pages for each AWS service you plan to use, for cost estimates.

This deployment requires a license for Data Quality. To sign up for a license, contact Informatica.

The following table lists the instance types that you can choose based on sizing requirements:

Virtual Machine	Instance Type
Database	M5.xlarge / R5.xlarge
Informatica domain	M4.2xlarge / M5.2xlarge

The deployment chooses M4 or M5 type of instances based on instance-type availability in a specific AWS region.

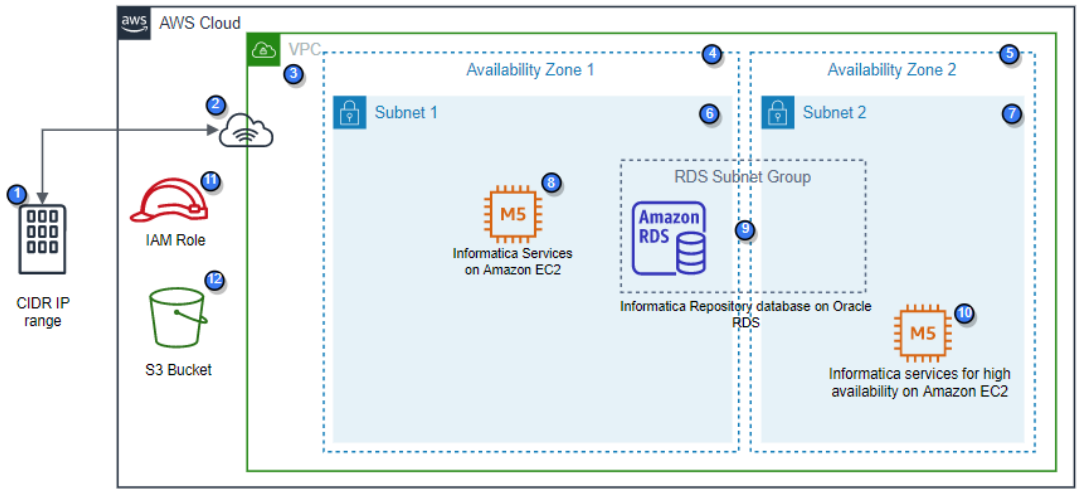
Apache Log4j RCE Vulnerability

Informatica 10.5.1.1 addresses the Log4j RCE vulnerabilities by removing the JndiLookup class from the vulnerable Log4j libraries.

For information about the vulnerabilities, see Informatica Response to Apache Log4j RCE Zero Day Vulnerability.

Architecture

The following image shows the architecture of Data Quality on AWS U.S. Intelligence Community Marketplace:



The architecture diagram includes the following components:

1. Your company or home network CIDR IP range.
2. A gateway to enter AWS VPC resources. It can be a NAT gateway, Internet gateway, or a virtual private gateway.
3. A virtual private cloud (VPC) configured in the Availability Zone to contain the Data Quality deployment.
4. Availability Zone 1. This deployment provisions one subnet.
5. Availability Zone 2. This deployment provisions one subnet.
6. Subnet 1 to contain specific elements of the deployment, such as Amazon EC2 and Amazon RDS databases. This creates one subnet.
7. Subnet 2 to contain specific elements of the deployment for high availability, such as Amazon EC2. This creates one subnet.
8. Informatica domain, including the Model Repository Service and the Data Integration Service.
9. Oracle database to contain the following Informatica repositories:
    a. Domain configuration repository
    b. Model repository, a relational database that stores all the metadata for projects created using Informatica client tools. The Model repository also stores run-time and configuration information for applications that are deployed to a Data Integration Service.

    You can use a single Oracle database for both repositories with a different schema for each repository.
10. The Informatica domain, including the Model Repository Service and the Data Integration Service for high availability.
11. IAM roles.
12. S3 storage to act as a location for the license key.

Components in a Deployment

The deployment creates the following components when you deploy the software in an existing virtual private cloud:

Component	Components Created
Security groups	The following security groups: - One security group for RDS. - One security group for the Informatica domain instance.
Database subnet group	1
Amazon RDS instance	One Amazon RDS for Oracle instance with 200 GB of storage capacity that runs on a db.m5.xlarge DB instance type.
Amazon EC2 instances	Informatica domain with an EC2 instance and 500 GB of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region. Based on your choice, 1 (one) Informatica high availability domain node on an EC2 instance with 500 GB of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region.

Informatica domain

The Informatica domain is a server component that hosts application services, such as the Model Repository Service and the Data Integration Service. These services, together with domain clients, enable you to create and run mappings and other objects to extract, transform, and load data.

Application Services

Model Repository Service

The Model Repository Service is an application service in the Informatica domain that manages the Model repository. The Model repository stores metadata created by Informatica products in a relational database to enable collaboration among the products. Informatica Developer, the Data Integration Service, and the Administrator tool store metadata in the Model repository.

Data Integration Service

The Data Integration Service is an application service in the Informatica domain that performs data integration tasks for the Developer tool and for external clients.

Content Management Service

The Content Management Service is an application service in the Informatica domain that manages reference data and is responsible for compiling rule specifications into mapplets. The Content Management Service provides reference data information to the Data Integration Service and to the Developer tool and Analyst tool. The Content Management Service stores reference data in a database that you specify.

Analyst Service

The Analyst Service is an application service in the Informatica domain that runs Informatica Analyst. The Analyst Service manages the connection between the service components and the users who log in to Analyst tool. You can perform column and rule profiling, manage scorecards, and manage bad records and duplicate records in the Analyst tool. The Analyst Service stores profiling, scorecarding, and bad and duplicate record data in databases that you specify.

The Informatica domain can run several other services. For more information about Informatica services, see the Informatica Application Service Guide.

Repositories

Informatica repositories, hosted on Oracle or Microsoft SQL Server databases, store metadata about domain objects. Informatica repositories include the following:

Domain configuration repository

The domain configuration repository stores configuration metadata about the Informatica domain. It also stores user privileges and permissions.

Model repository

The Model repository stores metadata for projects and folders and their contents, including all repository objects such as mappings and workflows. The repository also stores rules you apply during data preparation.

For more information about domain repositories, see the Informatica Application Service Guide.

Informatica clients

You can use different clients with Data Quality:

Administrator tool

The Administrator tool enables you to create and administer services, connections, and other domain objects.

Analyst tool

The Analyst tool enables you to analyze, cleanse, standardize, profile, and score data in your enterprise.

Developer tool

The Developer tool enables you to create and run mappings and other objects that enable you to access, transform, and write data to targets.

Command line interface

The command line interface offers hundreds of commands to assist in administering the Informatica domain, creating and running repository objects, administering security features, and maintaining domain repositories.

Before You Begin

Before you launch the automated deployment on the AWS U.S. Intelligence Community Marketplace, verify the prerequisites for each option described in this section.

Network Prerequisites - AWS

Before you deploy Data Quality, verify that you have completed the following network prerequisites for the Informatica application services.

When you deploy to an existing VPC, the automated deployment provisions Data Quality in your existing AWS infrastructure.

Existing VPC Prerequisites

Complete the following prerequisites when you choose to deploy into an existing VPC:

- Ensure that the VPC has DNS resolution enabled.
- You must also attach the subnet to a route table with a local route to the VPC CIDR.
- Verify that each subnet in the VPC is associated with a route table.

- Verify that each route listed in the route table is defined as one of the following gateway types:
  - Internet gateway
  - NAT gateway
  - Virtual private gateway
- Verify that the subnet is associated with one of the following gateways:
  - Internet gateway
  - NAT gateway
  - Virtual private gateway
- Verify that the subnets you choose are attached to a route table under the selected VPC. Ensure that S3, EMR, and Cloudformation endpoints are setup for the selected VPC and subnets.
- Verify that your VPC has a minimum of two subnets.
- Ensure that the EC2 user has the ListBucket, GetObject, and PutObject permissions on the RDS bucket.
- Ensure that the RDS role arn has the ListBucket and GetObject permissions and access permission to the selected keypair and the License key S3 bucket.
- Ensure that an IAM role and an IAM instance profile with the same name are present with ListBucket, GetObjects, and ListObjects permissions. While deploying, use the same bucket and folders that are part of this IAM role policy resources.
- If the selected subnets are private, make sure that they can communicate to the AWS services using VPC endpoints, NAT gateway, or any other available connectivity options.

Record the domain name configured in the DHCP options as explained in the Amazon VPC documentation. You provide the domain name in the VPC settings when you configure the deployment.

Account and Security Prerequisites

Complete the prerequisites in this section to verify that the AWS user has the required account user policies.

Account User Policies

Choose to grant the following privilege to the AWS user:

- Administrator privileges
- The following minimum policies attached to the user account:
  - arn:aws:iam::aws:policy/AWSCloudFormationFullAccess
  - arn:aws:iam::aws:policy/AmazonEC2FullAccess
  - arn:aws:iam::aws:policy/IAMFullAccess
  - arn:aws:iam::aws:policy/AmazonS3FullAccess
  - arn:aws:iam::aws:policy/AmazonRDSFullAccess
  - arn:aws:iam::aws:policy/AmazonVPCFullAccess

Storage Prerequisite

Create an Amazon S3 bucket for the Data Quality license.

Amazon S3 stores objects, consisting of one or more files and accompanying file metadata, in "buckets." For information about Amazon S3, see the AWS documentation.

To use the automated solution on the cloud marketplace, you must create an S3 bucket to store the license. You can store the license on the bucket level or in a folder within the S3 bucket.

Note the location of the license key and supply it when you configure the solution in the **License Key S3 Bucket URI** parameter.

Deploy Data Quality on the AWS U.S. Intelligence Community Marketplace

The automated deployment of Data Quality on AWS U.S. Intelligence Community Marketplace uses the AWS CloudFormation console to guide your choices and launch the solution deployment.

The deployment also lets you configure additional settings such as CIDR blocks and Data Quality settings, as discussed later in this guide.

Use the AWS CloudFormation console to specify the stack name and to enter parameter values to the template that you use to deploy the application. A stack is a logical representation of an AWS deployment.

Specify the identifiers for the VPC, for the subnet that contains the Amazon EC2 instances where the deployment process deploys the application components, and for the subnet where the deployment process creates the Model repository. The deployment process provisions the application components and creates the Model repository based on the values that you specify.

Availability zones and subnets

The automated solution deploys the components and services for Data Quality into an Amazon VPC configured across two Availability Zones. Each Availability Zone is provisioned with two subnets.

In the first subnet, the deployment process provisions Data Quality application services on an Amazon EC2 instance within the VPC:

- Data Integration Service
- Model Repository Service
- Content Management Service
- Analyst Service

For more information about domain components, see the Informatica Administrator Guide and Informatica Application Service Guide.

In the second subnet, the solution deploys an Oracle or Microsoft SQL Server database on Amazon RDS for use as the Model repository. The Model repository stores metadata created by Informatica services.

Note: The same RDS instance stores metadata for the Informatica domain.

Deployment steps

You perform multiple tasks to deploy the Data Quality 10.5 AWS U.S. Intelligence Community Marketplace solution.

You configure multiple parameters, such as availability zones and subnets.

Step 1. Prepare an AWS account

Perform the following steps to prepare an AWS account:

1. If you do not already have an AWS account, create an account at aws.amazon.com.
2. Use the region selector in the navigation bar to choose the AWS region where you want to deploy Data Quality on AWS.

   Note: The region you choose determines the availability of certain sizes of EC2 instances.
3. Create a key pair in your preferred region.

4. When you log in to an EC2 instance, you use a password file for authentication. The file is called a private key file and has a file name extension of .pem.
5. If you do not have an existing .pem key to use, follow the instructions in the AWS documentation to create a key pair.
   When you create a key pair, you save the .pem file to your desktop system. Simultaneously, AWS saves the key pair to your account. Make a note of the key pair that you want to use for the Data Quality instance, so that you can provide the key pair name during network configuration.

   Note: Your administrator might ask you to use an existing key pair.
6. If necessary, request a service limit increase for the EC2 instance type that you want to use for the Informatica domain. You might need to do this if an existing deployment uses this instance type and you think you might exceed the default limit with this reference deployment.

Step 2. Get a license for Data Quality

This deployment requires a license for Data Quality. To sign up for a license, contact Informatica.

1. In your AWS account, create an S3 bucket.
2. Optionally, create a directory under the S3 bucket.
3. Place the Data Quality license key file for the software in the S3 bucket or directory.

Step 3. Launch the deployment

You are responsible for the cost of the AWS services used while running this deployment. There is no additional cost for using this deployment. See the pricing pages for each AWS service you will be using for full details. Each deployment takes approximately one to two hours to complete.

Perform the following steps to launch the deployment:

1. Log in to AWS U.S. Intelligence Community Marketplace using the AWS account.
2. Type Data Quality in the search field and press **Enter**.
3. Click **Data Quality**. The **Product Overview** page appears.
4. Click **Continue to Subscribe**. The **Subscribe to this software** page appears.
5. Click **Continue to Configuration**. The **Configure this software** page appears.
6. Select **Data Quality** from the **Delivery Method** list.
7. Choose **10.5.1.1** from the **Software Version** list.
8. Click **Continue to Launch**. The **Launch this software** page appears.
9. Select **Launch CloudFormation** from the **Choose Action** drop-down list and click **Launch**. The **Create stack** page appears.
10. On the **Specify template** section, keep the default setting for the template URL, and then choose **Next**.

    The default option is **Amazon S3 URL**.
11. On the **Specify stack details** page, change the stack name if needed. Review the parameters for the template. Provide values for the parameters that require input.
12. Under **Capabilities**, select the check box to acknowledge that the template will create IAM resources.
13. To deploy the stack, click **Create**.

    Amazon AWS begins to create the stack and displays the CloudFormation dashboard. When the status is CREATE_COMPLETE, the deployment is complete. You can use the URL displayed on the **Output tab** to view the stack properties.

14. For all other parameters, review the default settings and customize them as necessary. When you finish reviewing and customizing the parameters, choose **Next.**

    Note: Make sure that the VPC has at least two subnets.

Parameters for deployment

1. On the **Create Stack** page, enter a name for the stack. The stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).
2. Enter the parameters required to deploy the application.
   The following table lists the parameters in the **Network Configuration** section:

Parameter	Default	Description
VPC	Requires input	ID of your existing VPC where you want to deploy Data Quality.
Subnet 1	Requires input	Select the subnet ID where you want to deploy Data Quality. Make sure that the subnet is included in the VPC. Ensure that service subnet ID1 and service subnet are not the same.
Subnet 2	Requires input	If you want to deploy Data Quality with high availability enabled, provide the subnet ID that you want to use for high availability. Make sure that subnet 1 and subnet 2 are in different availability zones.
CIDR IP Range	Requires input	The CIDR IP range that is permitted to access Data Quality. To reduce the potential of inbound attacks from unknown IP addresses, you can use a constrained CIDR range. For example, to specify the range of 10.20.30.40 to 10.20.30.47, enter 10.20.30.40/29.

The following table lists the parameters in the **Amazon EC2 Configuration** section:

Parameter label (name)	Default	Description
Instance Profile	Requires input	Name of an existing instance profile with the same name as the IAM role to be attached to the EC2 instance. The IAM role must have the ListBucket, GetObject and PutObject permissions on the RDS bucket. The user should also have access to the selected keypair and the License key S3 bucket with GetObject and ListBucket permissions.
Key Pair Name	Requires input	Name of an existing Amazon EC2 key pair to enable SSH and Remote Desktop Protocol (RDP) access to Data Quality.
Keypair S3 URI	Requires input	S3 Bucket URI for the keypair selected. Use the following format: `s3://bucket-name/Keypair.pem` or `s3://bucket-name/sub-folder/Keypair.pem`

Parameter label (name)	Default	Description
RDS Role Arn	Requires input	Valid IAM role arn for RDS to access data from the RDS bucket. The RDS role arn should have access to the RDS bucket with ListBucket and GetObject permissions.
RDS Bucket	Requires input	Bucket RDS will uses this to retrieve data.

The following table lists the parameters in the **Informatica Data Quality Configuration** section:

Parameter label	Default	Description
License Key S3 Bucket URI	Requires input	The S3 bucket URI for the Data Quality license key in the following format: `s3://<bucket-name>/<directory>/<license key>`
High Availability Deployment	Requires input	Enable or disable high availability for the deployment. For information about high availability for the Informatica domain, see the Informatica documentation.

3. Click **Next**.
4. On the **Configure stack options** page, you can specify key-value pairs for resources in your stack and set advanced options. When you are done, choose **Next**.
5. On the **Review** page, review and confirm the template settings. Under Capabilities, select the check box to acknowledge that the template will create IAM resources.
6. Choose **Create** to deploy the stack.
7. Monitor the status of the stack. When the status is `CREATE_COMPLETE`, the deployment is complete.

See the Data Quality Getting Started Guide for information about logging in to Data Quality.

Monitoring Instance Provision and Informatica Domain Creation

You can use cloud platform dashboards, logs, or other artifacts to see whether cluster creation succeeded and how to locate and identify the Informatica domain and Data Quality on the cloud platform.

Deployed AWS Resources

You can monitor the progress of the solution deployment in the Create Stack section of the AWS portal. Each row of the display shows stack components and creation status. The stack is complete when the status reads CREATE_COMPLETE.

When you deploy the solution to an existing VPC, the console displays the following stack components:

Component	Description
StackName/IDQStack	The main stack. Created by default.
StackName-SecurityGroupStack	Creates security groups and network interfaces used by EC2 instances. Created by default.

Component	Description
StackName-RDSStack	Creates RDS resources. Created by default.
StackName-IDQServerStack	Creates an Informatica gateway node on which Data Quality runs on an EC2 instance. Created by default.
StackName-IDQHANodeStack	Creates a high availability node on which Data Quality runs on an EC2 instance. Created when you set High Availability Deployment to Enabled.

Output Tab Properties

Click the **Output** tab to view the list of resources that the deployment process created in the VPC. The following table lists the outputs:

Output	Description
AdminConsoleUser	User name of the Administrator tool user.
AdminConsoleUserPwd	Password for the Administrator tool user.
DBUserPwd	Password for the database users of the domain databases created in the deployment.
DBUsers	User IDs for the database users of the domain databases created in the deployment.
HANodeIP	IPv4 address of the high availability or failover node for the domain.
InformaticaAdminConsolePrivateURL	URL:port of the Administrator tool hosted on a private server.
InformaticaAdminConsoleServerLog	Path to the domain installation log.
InstanceSetupLogs	Path to the domain EC2 host setup log.
InformaticaAnalystPrivateURL	Informatica Analyst tool URL (Private DNS)

Use the URL, user ID, and password to access the Administrator tool to perform administrator tasks.

Deployed Informatica Resources

The list of deployed resources includes the **InformaticaAdminConsolePrivateURL**.

To access the Administrator tool, use the URL in the following format:

<Private IP address>:8443

The Administrator tool opens in a browser window.

Use the Informatica Administrator user name and password listed in the **Outputs** list to log in to the Administrator tool.

You can use the Administrator tool to view and administer the Informatica application services and resources that the automated deployment created.

Change default password

▶ Before you start using the product, change the following default passwords:

Application	User name	Default password	How to change
Informatica Administrator	Administrator	Administrator@1	See the Changing your password topic.
DBUserPwd	admin	infa_12345	See the Changing your password topic.

Solution Deployment Log Files

You can review the following log files to see the solution deployment as it happened.

Review the following log files in the `/var/log` directory:

- cfn-init.log
- cfn-init-cmd.log
- cfn-wire.log
- cloud-init.log
- messages

Review the following log file in the `/opt` directory:

- Infa_OneClick_Solution.log

Informatica Domain Logs

You can access the following logs on the Informatica domain VM:

Command execution log

This log records the following events:

- Creation of Informatica connections, cluster configurations, and services.
- Population of the data warehouse and SQL databases.
- Data Integration Service recycling to register all changes.

At the top of the log file is a summary section that lists automated tasks and their status. Beneath the summary section are detailed sections about each task. If any of the tasks failed to complete successfully, you can look at the detailed section for the task to troubleshoot the task.

Find the following files in the /opt directory on the Linux machine:

- Infa_OneClick_Solution.log
- Infa_node_configure.log

Find the following files in the /var/log directory on the Linux machine:

- cfn-init-cmd.log
- cfn-init.log
- cloud-init.log

Troubleshooting

Review the AWS log files to troubleshoot issues with your Data Quality deployment.

Stack failure

Review the following log files if an EC2 instance where Informatica services run is running but the stack fails, or if the Administrator tool fails to launch:

- Node preparation log file: `/opt/Infa_node_configure.log`
- Installation log file: `/opt/Infa_OnceClick_Solution.log`
- Command Initialization log file: `/var/log/cfn-init-cmd.log`
- Service Initialization log file: `/var/log/cfn-init.log`
- Cloud log file: `/var/log/cloud-init.log`

Frequently asked questions on deployment error messages

I encountered a CREATE_FAILED error when I launched the deployment.

If AWS CloudFormation fails to create the stack, we recommend that you relaunch the template with **Rollback on failure** set to **No**. (This setting is under **Advanced** in the AWS CloudFormation console, **Options** page.) With this setting, the stack's state will be retained, and the instance will be left running, so you can troubleshoot the issue. (You will want to look at the log files in `/var/log/cfn-init-cmd.log`)

Note: When you set **Rollback on failure** to **No**, you will continue to incur AWS charges for this stack. Please make sure to delete the stack when you have finished troubleshooting.

For additional information, see Troubleshooting AWS CloudFormation on the AWS website or contact us on the AWS Discussion Forum.

I encountered errors while installing the Informatica domain and services. Where can I find more information?

View the `/opt/Infa_OneClick_Solution.log` log file to see more information about the errors that you encountered.

I encountered a size limitation error when I deployed the AWS CloudFormation templates.

We recommend that you launch the deployment templates from the location we provided or from another S3 bucket. If you deploy the templates from a local copy on your computer or from a non-S3 location, you might encounter template size limitations when you create the stack. For more information about AWS CloudFormation limits, see the AWS documentation.

What should I do before I relaunch the template with Rollback on failure?

Ensure that you disable rollback when you launch the stack in case of a stack failure. If Data Quality server stack is created, check the following logs to identify the issues:

- `/opt/Infa_OneClick_Solution.log`
- `/var/log/cfn-init.log`

I am facing Model Repository Service connectivity issues with the following error messages:

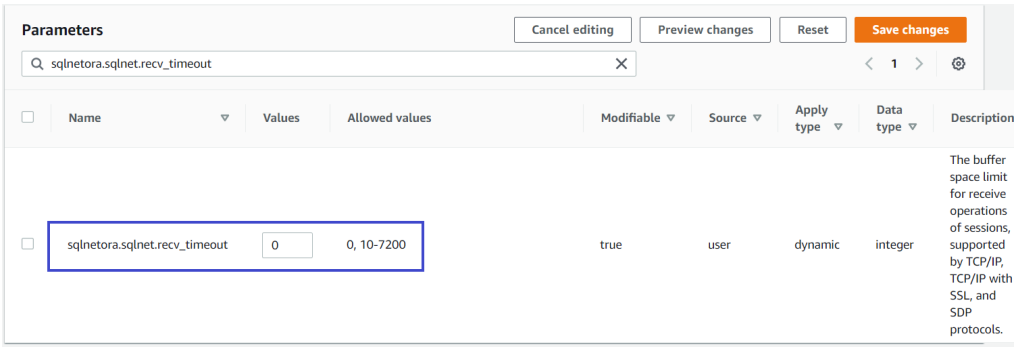
```
MRS_50015 "The Repository Service operation failed. ['[PRSVCSHARED_01707] Connection
issues with the configured database. Request failed with the error message
[[PERSISTENCEAPI_0307] [DBPERSISTER_1005] Failed to process requested operation. This
was caused by [informatica][Oracle JDBC Driver]No more data available to read.].']"
```

```
MRS_50015 "The Repository Service operation failed. ['[PRSVCSHARED_01707] Connection
issues with the configured database. Request failed with the error message
[[PERSISTENCEAPI_0307] [PERSISTENCECOMMON_0001] Internal error. The request processing
failed. This was caused by [informatica][Oracle JDBC Driver]Exception generated during
deferred local transaction handling.  See next exception via
SQLException.getNextException for details.].']"
```

Perform the following steps to resolve the Model repository database connectivity issue:

1. Launch the AWS RDS page.
2. Select the RDS instance.
3. Click the **Configuration** tab.
4. Scroll down the tab page and click the link under the **Parameter group** section.
5. Search for the following parameters in the list of parameters that appear:
   - sqlnetora.sqlnet.send_timeout
   - sqlnetora.sqlnet.recv_timeout
   - sqlnetora.sqlnet.inbound_connect_timeout
   - sqlnetora.sqlnet.outbound_connect_timeout
   - sqlnetora.tcp.connect_timeout

   Note: If the listed parameters are not present in the list, make sure that add the parameters to the list.
6. Change the value of each of the parameters listed in the previous step to 0 as shown in the following sample image:
   The following image shows the `sqlnetora.sqlnet.recv_timeout` parameters



7. Click **Save changes**.

Additional Resources

AWS Services

- AWS CloudFormation
- Amazon EC2
- Amazon VPC
- Amazon RDS
- Amazon S3
- Elastic IP Addresses

If you are new to AWS, see Getting Started with AWS.

Data Quality Documentation

- Product Documentation

Reference Deployments

- AWS Deployment home page

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